
```
clc;
clear;

% Prompt user for the number of variables and equations
m = input('Enter no. of variables: ');
n = input('Enter no. of equations: ');

A = zeros(m, n);
B = zeros(m, 1);

% Input coefficients for equations
for i = 1:m
    for j = 1:n
        A(i, j) = input(sprintf('Enter coefficients for equation %d%d: ', i,
j));
    end
    B(i, 1) = input(sprintf('Enter the value b%d: ', i));
end

% Define symbolic variables
symsVars = sym('x', [1, n]);
disp('The symbolic variables are:');
disp(symsVars);

% Calculate ranks
rank_A = rank(A);
rank_A_B = rank([A B]);

% Check if B is a zero vector
if all(B == 0) % Homogeneous system
    if rank_A == m
        disp('The system is homogeneous and has a trivial solution.');
```

```
    else
        disp('The system is homogeneous and has non-trivial solutions.');
```

```
    end
else % Non-homogeneous system
    if rank_A == rank_A_B
        if rank_A < m
            disp('The system is non-homogeneous and has infinite solutions.');
```

```
        else
            disp('The system is non-homogeneous and has a unique solution.');
```

```
        end
    else
        disp('The system is non-homogeneous and has no solution.');
```

```
    end
end

% Solve the system if applicable
if rank_A == rank_A_B
    equations = arrayfun(@(i) sum(A(i, :) .* symsVars) == B(i), 1:m,
'UniformOutput', false);
    sol = solve([equations{:}], symsVars, 'ReturnConditions', true);
```

```
    for i = 1:n
        varName = char(symsVars(i));
        fprintf('Solution for %s: %s \n', varName, char(sol.(varName)));
    end
end
```

```
Error using input
Cannot call INPUT from EVALC.
```

```
Error in code3 (line 5)
m = input('Enter no. of variables: ');
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

Published with MATLAB® R2025b